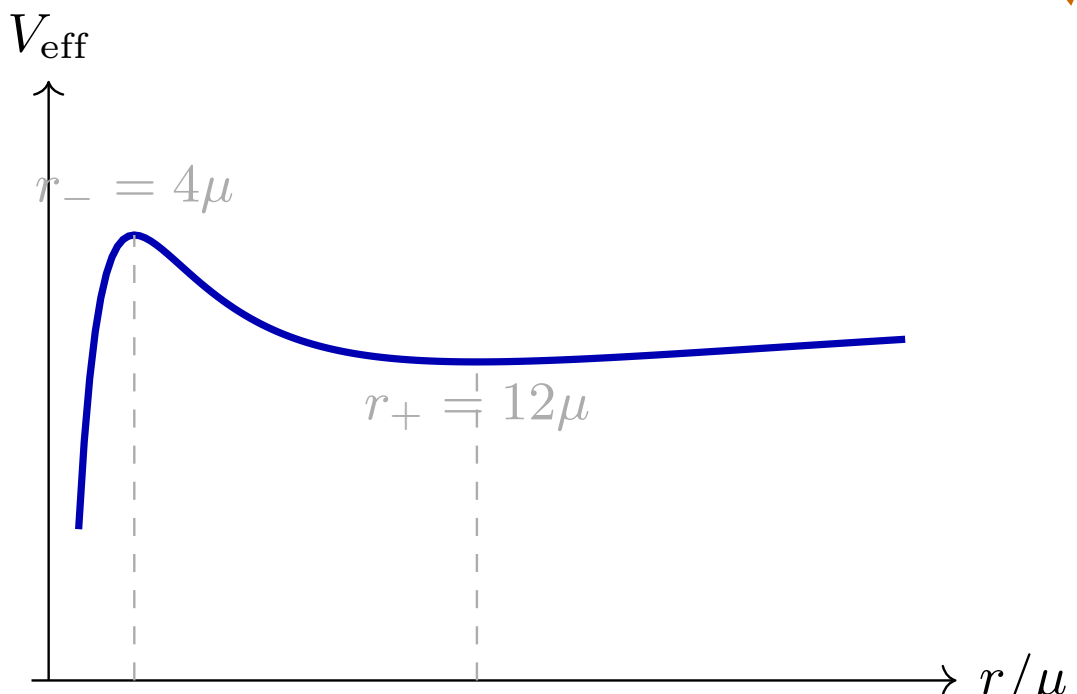
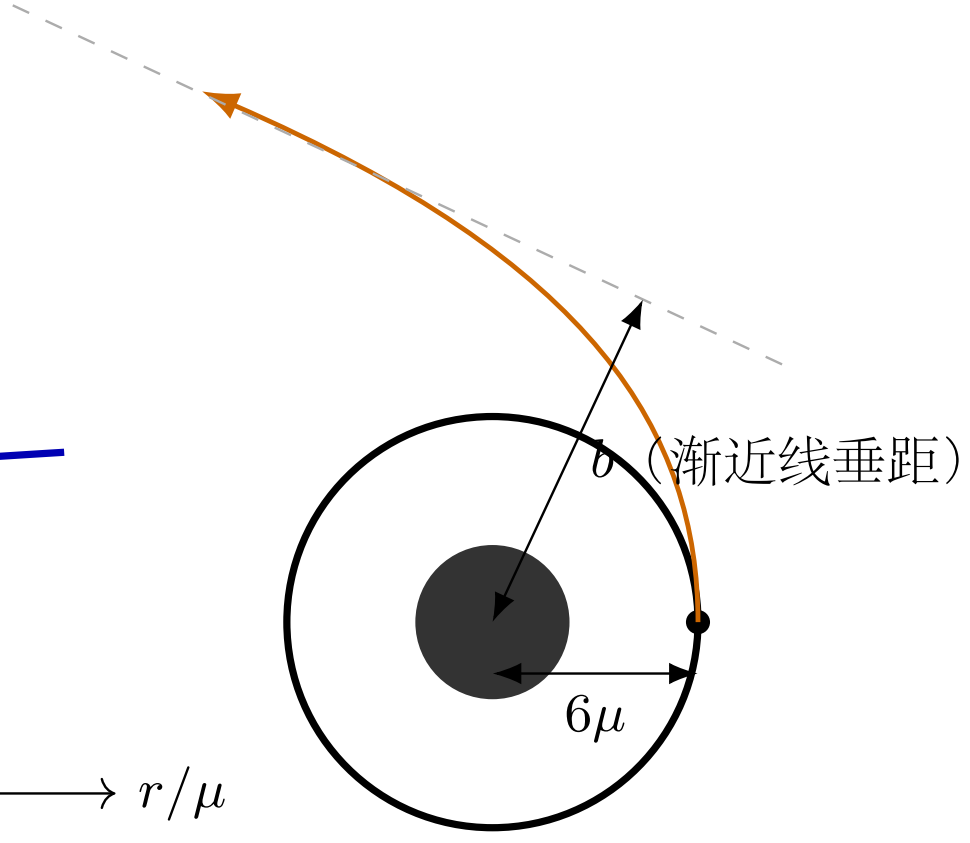




示例 $L = 4GMm_0/c$；稳定支 $r_+ > 6\mu$

光路示意；$\mu = GM/c^2$

$b = 3\sqrt{6}\mu, \quad \nu_\infty = \sqrt{2}\nu$